

Kernel / User Address Space Overview

0x0000_0000_0000_0000 → 0x0000_7FFF_FFFF_FFFF : User space
UTZERO (text) ... USTKTOP (stack/TOS)

KZERO identity map (early boot alias of low phys)

KTZERO+ : Kernel ELF text/data, Mach blocks, kernel heap
Per-CPU Mach/TSS/GDT, Proc stacks+structs

HHDM (saved_limine_hhdm_offset): phys memory mirror

KMAP (2 MiB) & VMAP (1 TiB) device windows at 0xfffffe..